

Co₂ Assessment -1

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Course Name : Data structure

Course Code : CSA0307

- ① Insert the following elements 100, 200, 300, 400, 500 into the linked stack and pop the elements 100 and 200 from the linked stack. Show the linked structure before and after linked stack operations.

Linked Stack operations.

Step 1:- push Elements

Insert : 100, 200, 300, 400, 500

Since a stack follows LIFO (Last In, First out), the last inserted element becomes the top.

Stack after push:

Top
500
400
300
200
100
NULL

Step 2:- pop Elements

pop 500 first, then 400 (because stack removes from the top).

Stack after pop:

Top
300
200
100
NULL

The given question contains an error. In a stack, elements are removed according to the LIFO principle. Therefore, after inserting 100, 200, 300, 400, and 500 the correct popped elements are 500 and 400, not 100 and 200.

② An online food delivery service receives the following orders O_1, O_2, O_3, O_4 . While O_2 is being prepared, an express order O_5 arrives.

Questions

- Explain why a normal queue may not satisfy customer requirements.
- Recommend an appropriate queue implementation and justify your choice.

Online Food delivery Service.

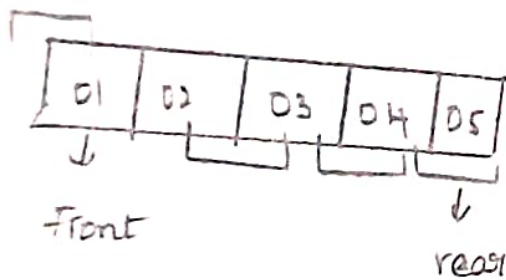
Given: orders received: O_1, O_2, O_3, O_4

While O_2 is being prepared, an Express order O_5 arrives.

(a) A Normal Queue follows the FIFO (First in, First out) principle. Therefore, the Express order O_5 is added at the end of the queue and has to wait until O_3 and O_4 are completed. This increases

the waiting time for the express order and may reduce customer satisfaction.

Normal Queue Diagram.

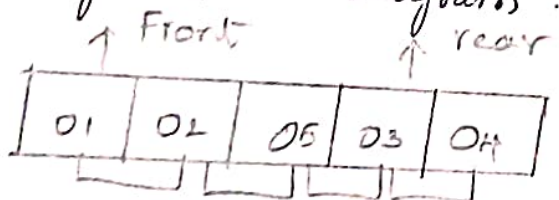


b) Recommended Queue: priority Queue

Justification:

- Express Orders are assigned higher priority
- O_5 is processed before O_3 and O_4
- It reduces waiting time and improves customer satisfaction.

priority Queue Diagram.



A priority Queue is the most suitable implementation because it allows express orders to be processed before normal orders, ensuring faster service and satisfaction.